



Cambridge IGCSE™

MATHEMATICS

0580/21

Paper 2 (Extended)

May/June 2020

MARK SCHEME

Maximum Mark: 70

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam. The answer column of the mark scheme shows the proposed basis on which Examiners would award marks for this exam. Where appropriate, this column also provides the most likely acceptable alternative responses expected from students. Examiners usually review the mark scheme after they have seen student responses and update the mark scheme if appropriate. In the June series, Examiners were unable to consider the acceptability of alternative responses, as there were no student responses to consider.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of 7 printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles

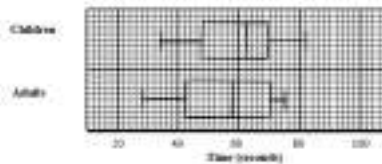
- | | |
|---|---|
| 1 | Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing. |
|---|---|

2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfw	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1	86	2	M1 for correct method to find the perimeter e.g. $(8 + 3) \times 2 + 5 + 3 + 8$ If 0 scored, SC1 for answer 98
2	15	1	
3(a)	66	1	
3(b)	Positive	1	
3(c)	Ruled line of best fit	1	
3(d)	46 to 50	1	FT <i>their</i> line of best fit if a positive gradient
4(a)	0.22 oe	2	M1 for $0.15 + 0.2 + ? + 0.43 = 1$ or better
4(b)	40	1	
5(a)	52	1	
5(b)	$7n + 5$ oe final answer	2	B1 for $7n + a$ or $bn + 5$ $b \neq 0$
6	7	3	M2 for $166 + 2x = 180$ or better or M1 for $97 - 3x + 69 + 5x = 180$ oe
7	$2^5 \times 3^4 \times 13^2$	1	
8	$\frac{56}{24} - \frac{21}{24}$	M2	M2 for correct method for common denominator or B1 for $\frac{7}{3}$
	<i>their</i> $\frac{35}{24} \times \frac{6}{25}$	M1	
	$\frac{7}{20}$	A1	
9(a)	$7a(3a + 4b)$ final answer	2	B1 for partial factorisation $7(3a^2 + 4ab)$ or $a(21a + 28b)$

Question	Answer	Marks	Partial Marks
9(b)	$5(2x + 3y)(2x - 3y)$ final answer	3	B2 for $(2x + 3y)(2x - 3y)$ or $(10x + 15y)(2x - 3y)$ or $(2x + 3y)(10x - 15y)$ or B1 for $5(4x^2 - 9y^2)$
10	$[x =] 55$ $[y =] 24$	2	B1 for each
11	990	3	M2 for correct complete area statement e.g. $\frac{1}{2} \times 30 \times (6 + 12) + 60 \times 12$ oe or M1 for one area calculation
12(a)	22	2	B1 for 48 and 70
12(b)		2	M1 for a box with two whiskers and at least two correct from Min 28, LQ 42, Med 58, UQ 70, Max 75
13	16.6 or 16.64...	5	M2 for $21 \times \frac{18}{13.5} = [AC]$ oe or M1 for scale factor $\frac{13.5}{18}$ or $\frac{18}{13.5}$ oe soi Then Pythagoras method: and M2 for $\sqrt{28^2 + 18^2} [\div 2]$ or $\sqrt{(theirAC)^2 + 18^2} [\div 2]$ or M1 for $AD^2 = 28^2 + 18^2$ or $AD^2 = (theirAC)^2 + 18^2$ OR alternative trigonometry method e.g. M1 for $\tan E = \frac{21}{13.5}$ and M1 for $AD = \frac{18}{\cos their 57.3}$
14(a)	$[p =] -13$	2	M1 for $4(5x - 4) + 3$ or better
14(b)	$\frac{3x+1}{5}$	3	M2 for $x = \frac{3y+1}{5}$, $5y = 3x + 1$ or $y - \frac{1}{5} = \frac{3x}{5}$ M1 for $x = \frac{5y-1}{3}$, $3y = 5x - 1$ or $y + \frac{1}{3} = \frac{5x}{3}$

Question	Answer	Marks	Partial Marks
15	Complete explanation with geometrical reasons	3	B1 for $RQP = x^\circ$ QR bisects angle PQB B1 for $RPQ = x^\circ$ alternate segment theorem B1 for triangle PQR has two equal angles both less than 60 (so can't be equilateral) so must be isosceles
16	1.8 or $1\frac{4}{5}$	3	M2 for $m = \frac{k}{(p-1)^2}$ or M1 for $m = \frac{their k}{(6-1)^2}$ OR M2 for $5(4-1)^2 = m(6-1)^2$ oe
17(a)(i)	$\begin{pmatrix} 15 \\ 21 \end{pmatrix}$	1	
17(a)(ii)	26	2	M1 for $10^2 + (-24)^2$ or better
17(b)	$\mathbf{p} + \frac{3}{4} \mathbf{q}$	2	M1 for a correct route or for $\overrightarrow{AE} = \frac{3}{4} \mathbf{q}$
18	34	2	M1 for $12 + 0.5$ or $4 + 0.5$ or better seen
19	12.2 or 12.24...	5	M4 for $\tan = \frac{4.5}{\sqrt{20^2 + 5.5^2}}$ oe or M1 for recognising angle GAC M1 for $\frac{495}{20 \times 5.5}$ M1 for $\sqrt{20^2 + 5.5^2}$ or $\sqrt{20^2 + 5.5^2 + (their 4.5)^2}$ M1 for $\tan = \frac{their 4.5}{\sqrt{20^2 + 5.5^2}}$ oe
20	$[y =] 5x - 4$	1	
21	$3x^3 - 7x^2 - 43x + 15$	3	B2 for correct expansion and simplification of two of the brackets or B1 for correct expansion of two brackets with at least 3 terms correct

Question	Answer	Marks	Partial Marks
22	142 or 142.2 to 142.3	3	M2 for $\frac{1}{2} \times 7.4 \times 7.4 \times \sin 60 \times 6$ or $\tan 60 \times \frac{7.4}{2} \times \frac{7.4}{2} \times 6$ or M1 for $\frac{1}{2} \times 7.4 \times 7.4 \times \sin 60$ or $\tan 60 \times \frac{7.4}{2}$